

PRINCY MAHESHWARI

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Education

Georgia State University

Bachelor of Science in Computer Science

GPA: 4.3 / 4.0

Presidential Scholar – Full-ride merit scholarship awarded to top two international students

May 2029

Atlanta, GA

Research Experience

Math Path Research Scholar – Data Mining Lab (DMLab) GSU

Oct 2025 – Present

Advisor: Prof. Berkay Aydin

Atlanta, GA

- Collected, cleaned, and integrated large-scale satellite datasets from NASA GOES and NOAA ACE missions to analyze solar flare activity and space-weather indicators.
- Built Python-based data processing pipelines using Pandas and NumPy to handle missing values, normalize telemetry streams, and prepare structured datasets for analysis.
- Performed exploratory data analysis and statistical correlation studies to identify relationships between solar wind parameters and flare intensity.
- Developed baseline machine learning models using scikit-learn to forecast solar storm timing and magnitude, evaluating model performance using error metrics and validation splits.
- Translated analytical findings into visual insights to support early-warning research objectives.

Undergraduate Research Assistant – Department of Computer Science GSU

Aug 2025 – Present

Advisor: Prof. Hemanth Venkateswara

Atlanta, GA

- Currently completing the Deep Learning Specialization by Andrew Ng, focusing on the design and training of deep learning models for image generation and sequence-based data.
- Learning and implementing CNN-based architectures for spatial data and RNN-based models for time-series and sequential modeling with emphasis on architecture selection and hyperparameter tuning.
- Implemented and evaluated neural network models using TensorFlow and NumPy, contributing to early-stage research exploring weak supervision and scalable learning workflows.

Projects

ToolFinder – HackIllinois 2026

Mar 2026

- Built ToolFinder, a multimodal AI system that enables voice-driven object search in cluttered workspaces by combining real-time camera feeds with computer vision to detect and highlight tools on demand.
- Designed and trained a **custom YOLO object detection model using Roboflow**, achieving **87% mAP@50, 88.4% precision, and 82.6% recall** on a curated dataset of workshop tools; integrated **SAM2 segmentation (Meta AI)** to refine detections with pixel-level masks and compute centroids for precise object localization.
- Implemented a GPU-accelerated inference pipeline on Modal, orchestrating speech transcription, semantic query mapping, and parallel vision models to return annotated detections and spatial coordinates for real-time object highlighting.

Veritas.dev – NexHacks

Jan 2026

- Built an automated GitHub App that detects mismatches between source code and documentation by ingesting repository changes via webhooks, parsing multi-language codebases, and flagging inconsistencies for developer review.
- Designed and implemented Python-based backend pipelines to normalize parsed code and documentation into structured representations, enabling anomaly detection and validation of parameters, types, and API signatures across real-world repositories.
- Integrated backend analysis with an interactive frontend dashboard to visualize trust scores, detected inconsistencies, and verification results, supporting reproducible workflows through GitHub issues, version-controlled outputs, and iterative review of detected inconsistencies.

NutriGirl AI – HackHer GSU 2025 (2nd Overall, 2nd Health Track)

Oct 2025

- Designed an AI-powered platform to help women manage PCOS/PCOD by combining lab report interpretation, nutrition tracking, and pose-aware workout coaching in a private, women-first interface.
- Led backend and computer vision development, building a smart lab analyzer for instant blood report summaries and a GPT-Vision-based pose detection system for real-time workout form correction.
- Integrated OpenAI APIs, Google MediaPipe, Web Speech API, and Node.js/Express.js to enable voice-guided feedback, automated PDF health reports, and a 24/7 chatbot trained on PCOS-specific care data.

Technical Skills

Languages: Python, JavaScript, HTML/CSS

Frameworks & Libraries: TensorFlow, PyTorch, NumPy, Scikit-learn, matplotlib, React.js, Node.js